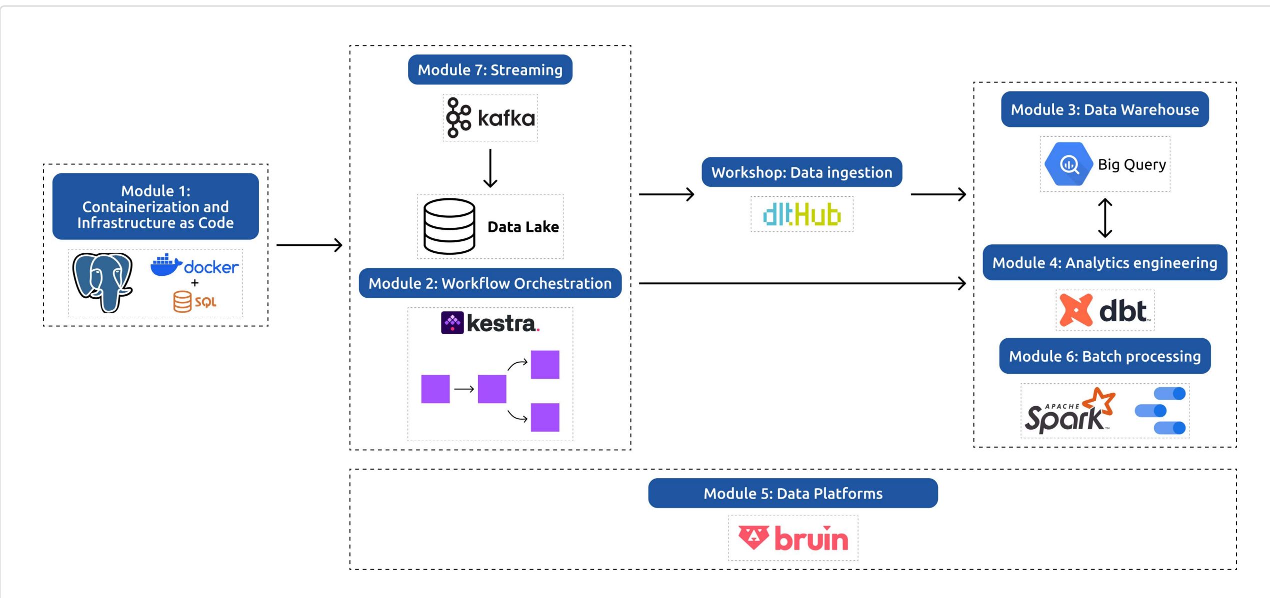
Data Engineering Zoomcamp: Free Data Engineering Course and Certification

Become a Data Engineer: Master Modern Data Engineering with Hands-On Training

11 May 2026 by Valeriia Kuka

Breaking into data engineering takes real, hands-on experience with production tools, but most courses stop at theory.

The Data Engineering Zoomcamp changes that. It's a free data engineering course that teaches you how to build production-grade data pipelines from start to finish. You'll work with Docker, Terraform, BigQuery, dbt, Spark, and Kafka, and graduate with a portfolio project and a certificate.



Complete Data Engineering Zoomcamp curriculum: from infrastructure setup to stream processing

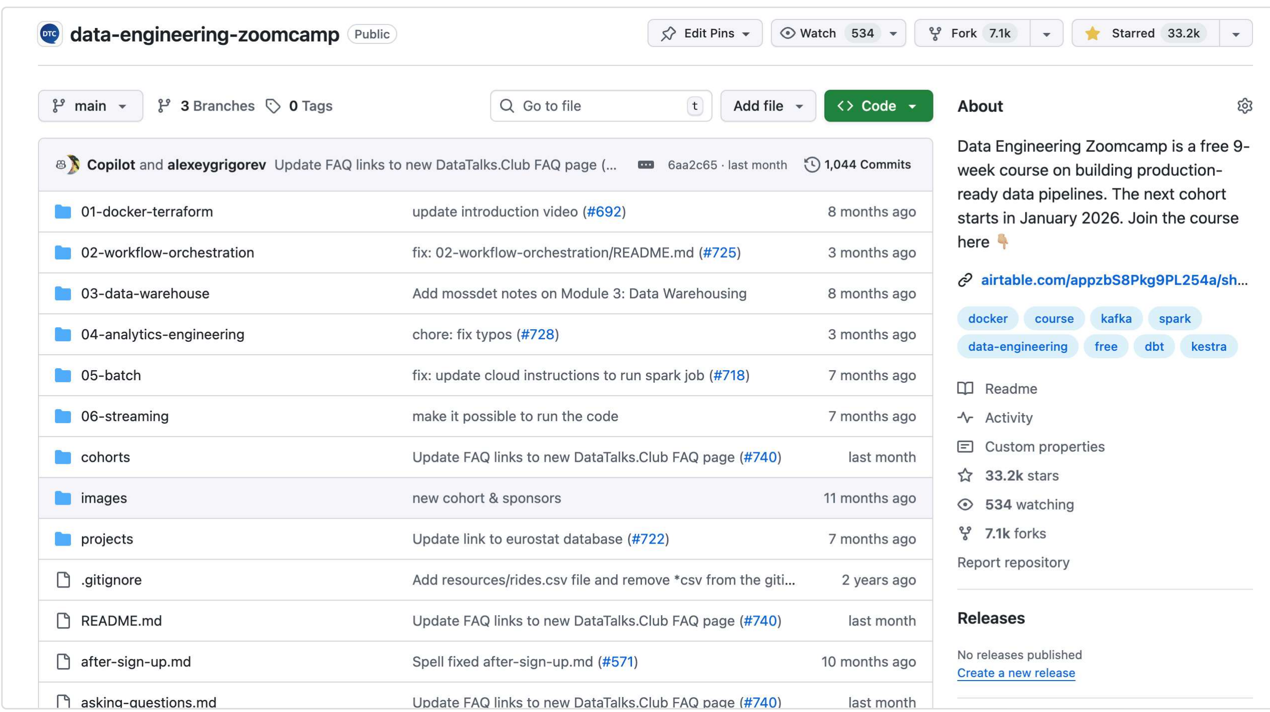
It's ideal for beginners and career switchers preparing for junior data engineer roles, as well as experienced professionals who want to refresh their knowledge, expand their network, or test themselves as a mentor for less experienced professionals.

Unlike most courses, DE Zoomcamp helps you build your public portfolio, share your work confidently, and connect with a global community for feedback, mentorship, and career opportunities.

TL;DR: Data Engineering Zoomcamp is a free 9-week course on building production-ready data pipelines. The next cohort starts in January 2026. Join the course here.

What is the Data Engineering Zoomcamp?

Data Engineering Zoomcamp is a 9-week program that follows a clear progression: infrastructure setup, workflow orchestration, data warehousing, analytics engineering, batch processing, streaming, and a final capstone project.



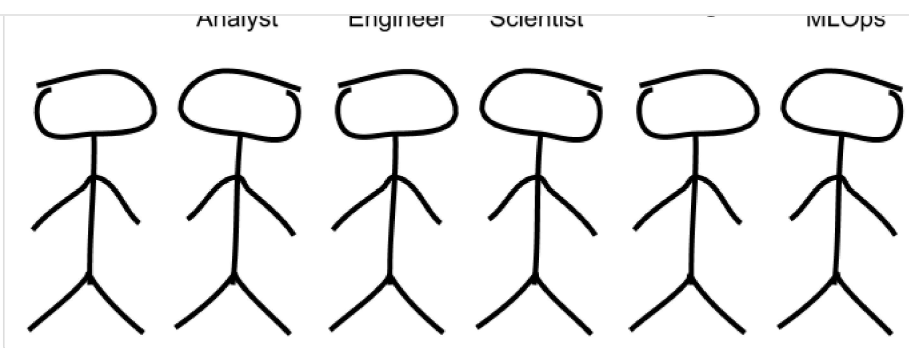
Data Engineering Zoomcamp GitHub repository showing the course materials

What makes it different is the community. You'll join an active Slack workspace where thousands of learners troubleshoot together, share progress, and connect for jobs and collaborations. The course encourages learning in public: sharing your work earns bonus points and builds your online presence.

The final three weeks focus on your capstone that gets peer-reviewed, and you graduate with a polished GitHub portfolio that proves you can ship real data systems.

Why Learn Data Engineering?

Typically, data science teams are comprised of data analysts, data scientists, and data engineers. Among different [data roles](#), data engineers are the specialists who connect all the pieces of the data ecosystem within a company or institution.



Data engineers are the specialists who connect all the pieces of the data ecosystem.

But why would you want to become one? Here are some of the main reasons that make data engineering roles rewarding and valuable:

1. You become the builder behind every data product that keeps information flowing.
2. You increase your earning potential by joining a smaller, high-value pool of professionals.
3. You develop transferable skills that are valuable across industries and provide long-term career flexibility. These skills are also foundational for roles in machine learning and MLOps.
4. You future-proof your career by building a mindset that will stay essential even as tools change and processes get automated.

Who Can Learn Data Engineering (and What Roles It Leads To)

The great thing about data engineering is that you don't need to be a computer science graduate or have years of experience in data to start.

If you understand basic programming and have curiosity about how data systems work, you can learn data engineering.

If you tick most of the following boxes, data engineering might be a good fit for you:

- Enjoy solving technical, logical problems and building things that work reliably.
- Like Python, SQL, or scripting and want to use those skills for something impactful.
- Want to understand how data moves inside organizations — from raw sources to analysis and AI (including LLM applications).
- Prefer practical, system-level work over purely theoretical or statistical modeling.
- Appreciate clear structure and step-by-step learning (which is how DE Zoomcamp is built).

Course Prerequisites

As mentioned earlier, learning data engineering doesn't require prior data engineering experience.

The course is designed to be accessible to beginners.

The only requirements are:

- Comfort with the command line (basic navigation and file operations)
- Basic SQL knowledge (SELECT, WHERE, JOIN statements)
- Python experience is helpful but not required

If you're completely new to programming, consider spending a few weeks learning the basics before starting the course.

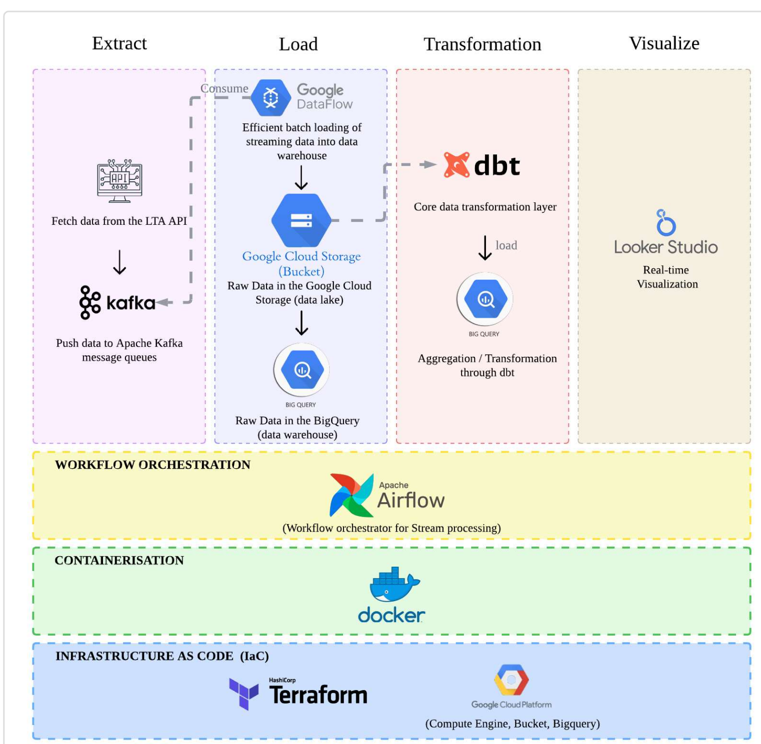
Course Curriculum: What You'll Learn in the Data Engineering Zoomcamp

The course follows a logical progression from infrastructure setup to advanced data processing, culminating in an end-to-end project.

1. Infrastructure & Prerequisites	· Set up your development environment with Docker and PostgreSQL · Learn cloud basics with GCP · Master infrastructure-as-code using Terraform	Docker, PostgreSQL, GCP, Terraform
2. Workflow Orchestration	· Master data pipeline orchestration with Kestra · Implement and manage Data Lakes using Google Cloud Storage · Build automated, reproducible workflows	Kestra, Google Cloud Storage
3. Data Warehouse	· Deep dive into BigQuery for enterprise data warehousing · Learn optimization techniques like partitioning and clustering · Implement best practices for data storage and retrieval	BigQuery
4. Analytics Engineering	· Transform raw data into analytics-ready models using dbt · Develop testing and documentation strategies · Create impactful visualizations with modern BI tools	dbt, BI tools
5. Data Platforms	· Building end-to-end data pipelines with Bruin · Data ingestion, transformation, and quality · Deployment to cloud (BigQuery)	Bruin, BigQuery
6. Batch Processing	· Introduction to Apache Spark · DataFrames and SQL · Internals of GroupBy and Joins	Apache Spark, Spark SQL
7. Streaming	· Introduction to Kafka · Kafka Streams and KSQL · Schema management with Avro	Kafka, Kafka Streams, KSQL, Avro
Final Project	· Build an end-to-end data pipeline from ingestion to visualization · Apply all learned concepts in a real-world project · Create a portfolio-ready project with documentation	Cloud platforms (GCP/AWS/Azure), Terraform, Spark, Kafka, dbt, BigQuery

Capstone Project

The final three weeks are dedicated to applying your knowledge in a real-world project that showcases everything you've learned throughout the course.



You'll build an end-to-end data pipeline using a dataset of your choice, implementing both data lake and warehouse solutions with proper documentation.

Your project will be peer-reviewed by fellow participants and you'll peer-review at least three other projects.

Project Requirements	**Deliverables**	**Evaluation Criteria**
· Select and process a dataset that interests you · Build end-to-end data pipelines (batch or streaming) · Implement both data lake and warehouse solutions · Create analytical dashboards	· Production-ready data pipeline · Documented data models · Interactive dashboard · Project presentation	· Peer review of at least three other projects · Technical implementation quality · Documentation completeness · Solution architecture design

Join the next cohort →

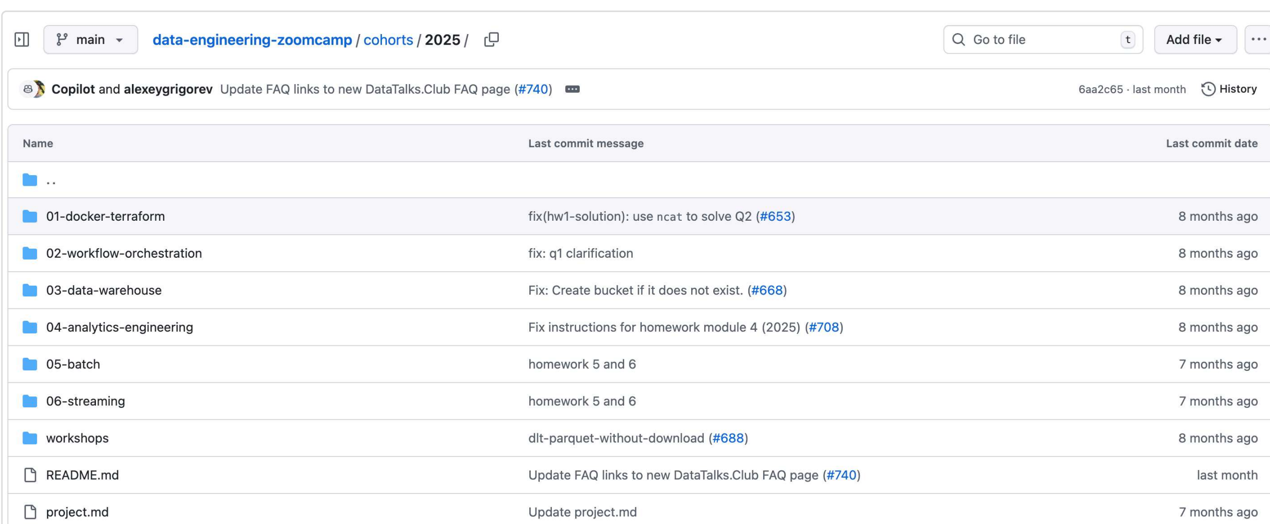
How the Data Engineering Zoomcamp Works

The course runs for 9 weeks in cohort format, providing structure and community support throughout your learning journey.

While you can access all course materials at your own pace without joining a cohort, participating in the structured program offers significant advantages: graded homework assignments, project submission and evaluation, peer interaction, and the opportunity to earn a certificate.

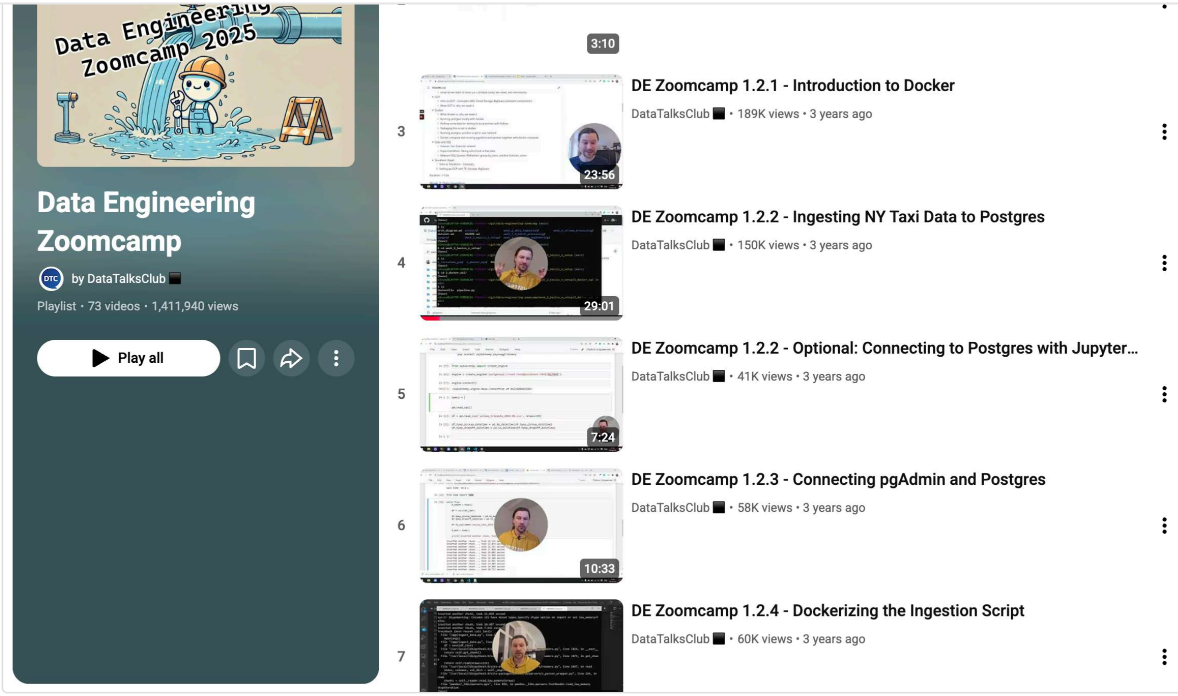
Below, we list the key features of the course and how they work.

GitHub Repository: The Central Hub of the Course



Data Engineering Zoomcamp GitHub repository showing the course materials

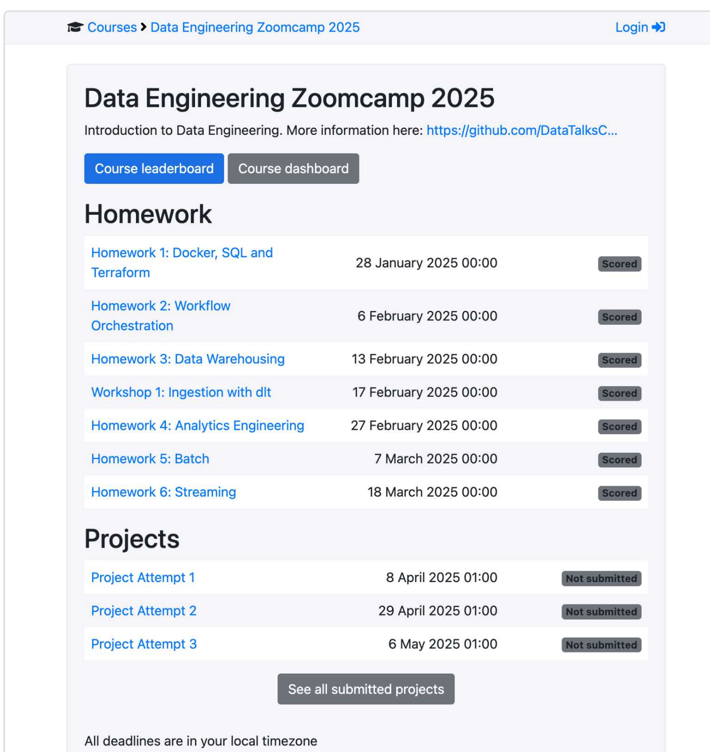
All course materials live in the GitHub repository. The lectures are pre-recorded and available on YouTube, so you can watch at your own pace.



Data Engineering Zoomcamp YouTube playlist with pre-recorded lectures

Homework Assignments

To reinforce your learning, we release homework assignments for each week of the course. You can submit a homework assignment at the end of each week.



Data Engineering Zoomcamp schedule showing the course schedule and submission deadlines

It doesn't count toward your certificate, but it helps you practice and appears on an optional anonymous leaderboard.

Your scores are added to an anonymous leaderboard, creating friendly competition among learners and motivating you to do your best.

722366d8b29ece9be3a7605363562c7c60d6918e	1	1	1	1	1	1	7	1	TRUE	14
636d3fa0090cfd4b22aeb4f3e7fe431274a2d6e8	1	1	1	1	1	1	7	1	TRUE	14
fbd7c94e3b9ad8a87aeac6839d98fb1de8e53a6e	1	1	0	1	1	1	7	1	TRUE	13
4ba1efa946abab7ea8b55ee35b8c1d9b2275dbc7	1	1	1	1	1	1	7	0	TRUE	13
d42547a15fc4d729c2f116231d22e28016a048ae	1	1	1	1	1	1	5	0	TRUE	11
b5674d8276208c9f2cdfc9d3511ab5132b5373da	1	1	1	1	1	1	5	0	TRUE	11
180e3b88144dcdeb95948091b18720b339b240a9	1	1	1	1	1	1	3	1	TRUE	10
f4d8ea9549ada13483c122cb792d176672040681	1	1	1	1	1	1	2	1	TRUE	9
d6bff240a8a416fa81018c4469c7fc8b1f544d24	1	1	1	1	1	1	2	1	TRUE	9
b6b5a3a7ca0dd3cd9748a777578f27436e712377	1	1	1	1	1	1	2	1	TRUE	9
acefbcffee893a9f9aac392088ec27845ab84322	1	1	1	0	1	1	3	1	TRUE	9
ab5250708f0086469885bac1f77602c00d955fa8	1	1	1	1	1	1	3	0	TRUE	9
5e1703f66746eeae783ab40f965f5f98a49864b2	1	1	1	1	1	1	3	0	TRUE	9
49c26a4774a99f0775aea77e3e34742fed76bce2	1	1	1	1	1	1	3	0	TRUE	9
3679a6319dcbf7f66a655a02f488e1d875168d85	1	1	1	1	1	1	2	1	TRUE	9
33895bfc211da2c395dcfc0e220791732a2c4979	1	1	1	1	1	1	3	0	TRUE	9
2c03147686d3670e5a5769514edb070ce0d11bca	1	1	1	1	1	1	2	1	TRUE	9
1ded918376ab51d806d6236c6e37f68b407e88ee	1	1	1	1	1	1	2	1	TRUE	9
fad7da9639425f5a2a1e3547c035cbeb85658303	1	1	1	1	1	1	1	1	TRUE	8
f8a886f14490cf630b462dfc8b1c0d06daa4c1d8	1	1	1	1	1	1	1	1	TRUE	8
f251db7236d9fd8a54a732d2870716d46eafb8d4	1	1	0	1	1	1	2	1	TRUE	8
e5dc14bef66a24181f85f6e2399005c75842e01e	1	1	1	1	1	1	1	1	TRUE	8
e5600efd4e2892305ce8b914ea1aacccce893f12	1	1	1	1	1	1	1	1	TRUE	8
e219c827723c25860e0811c0213cbe7d285d4e2a	1	1	1	1	1	1	1	1	TRUE	8
e217f84ee155fb6ee2cb7279f6768946a794b51f	1	1	1	1	1	1	1	1	TRUE	8
dc4cae2d0b4a8b831b318e09d71215134039409b	1	1	1	1	1	1	1	1	TRUE	8
d2b72d0fbe97ddc6f6abcb9259f408cb86f8e3f4	1	1	1	1	1	1	1	1	TRUE	8
ce7184efd2ca8b68b956a1a996a7f335d20eeefd	1	1	1	1	1	1	1	1	TRUE	8
cd0138161f7977242a8a47f814d05781613ec6af	1	1	1	1	1	1	1	1	TRUE	8
ccd22bee8eeca23c90c85a966d76730b17ee017a	1	1	1	1	1	1	1	1	TRUE	8
cbfa322d555de1bde5f140164183d5348c62e6ae	1	1	1	1	1	1	1	1	TRUE	8
c975da19a844797a66d984838ae04e5ba203ff84	1	1	1	1	1	1	1	1	TRUE	8
c90c3d72c306cb0c5ac9e97549cff4d92e752a65	1	1	1	1	1	1	2	0	TRUE	8

Course leaderboard displaying student progress and achievements anonymously

You can earn bonus points by learning in public — sharing your work on blogs, YouTube, or social media. The next section explains how it works in practice.

Learning in Public: Build Your Online Presence

A unique feature is our "learning in public" approach, inspired by Shawn @swyx Wang's article. We believe that everyone has something valuable to contribute, regardless of their expertise level.

> You already know that you will never be done learning. But most people "learn in private", and lurk. They consume content without creating any themselves. Again, that's fine, but we're here to talk about being in the top quintile. What you do here is to have **a habit of creating learning exhaust**:
>
> - Write blogs and tutorials and cheatsheets.
> - Speak at meetups and conferences.
> - Ask and answer things on Stackoverflow or Reddit. **Avoid** the walled gardens like Slack and Discord, they're not public.
> - Make Youtube videos or Twitch streams.
> - Start a newsletter.
> - Draw cartoons (people loooove cartoons!).
>
> Whatever your thing is, **make the thing you wish you had found** when you were learning. **Don't judge** your results by "claps" or retweets or stars or upvotes - just talk to yourself from 3 months ago. I keep an almost-daily dev blog written for no one else but me.

An extract from Shawn @swyx Wang's article about learning in public

Throughout the course, we actively encourage and incentivize learning in public. By sharing your progress, insights, and projects online, you earn additional points for your homework and projects.

DE Zoomcamp 2023 Leaderboard

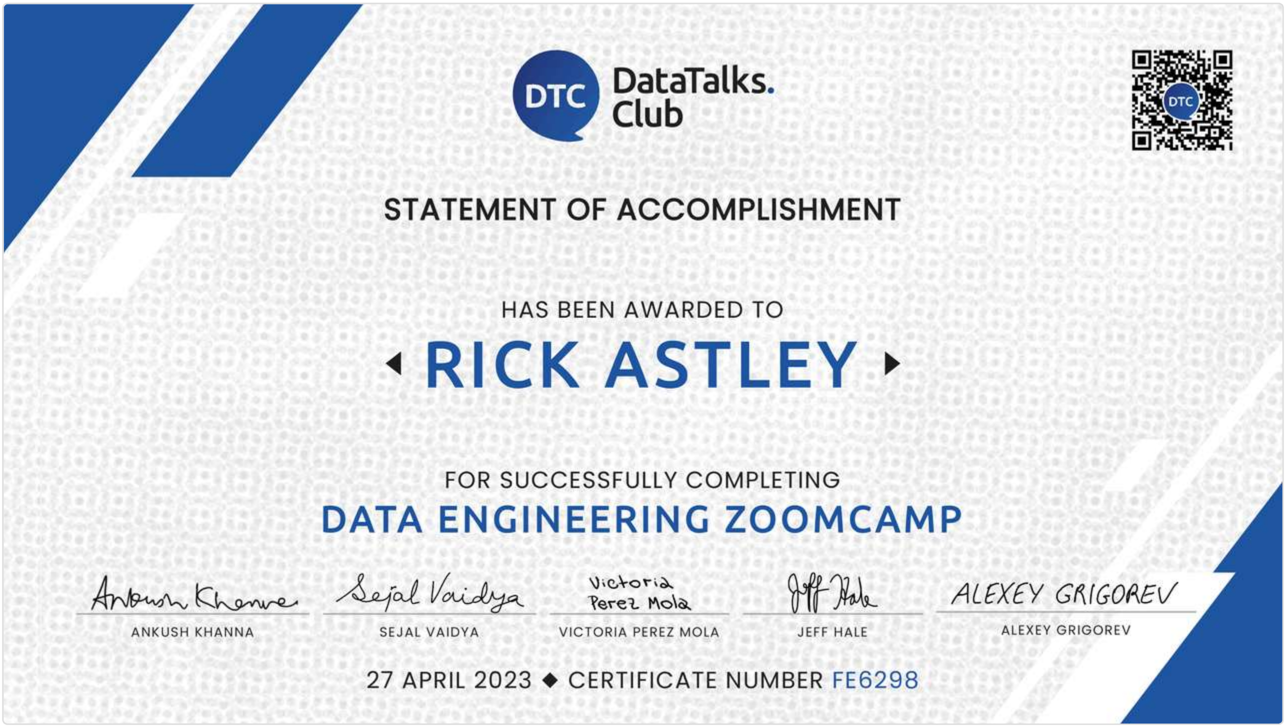
intro | hw-01a | hw-01b | hw-02 | hw-03 | hw-04 | hw-05 | hw-06 | hw-piperider | project-01 | project-02 | leaderboard

email	question1	question2	question3	question4	question5	question6	learning_in_publ	faq_score	valid_code_link	total_score
722366d8b29ece9be3a7605363562c7c60d6918e	1	1	1	1	1	1	7	1	TRUE	14
636d3fa0090cfd4b22aeb4f3e7fe431274a2d6e8	1	1	1	1	1	1	7	1	TRUE	14
fbd7c94e3b9ad8a87aeac6839d98fb1de8e53a6e	1	1	0	1	1	1	7	1	TRUE	13
4ba1efa946abab7ea8b55ee35b8c1d9b2275dbc7	1	1	1	1	1	1	7	0	TRUE	13

This not only demonstrates your knowledge but also builds a portfolio of valuable content. Sharing your work online also helps you get noticed by social media algorithms, reaching a broader audience and creating opportunities to connect with individuals and organizations you may not have encountered otherwise.

Many of our graduates have shared that their social media presence has helped them attract job offers and collaborations.

How to Get a Certificate

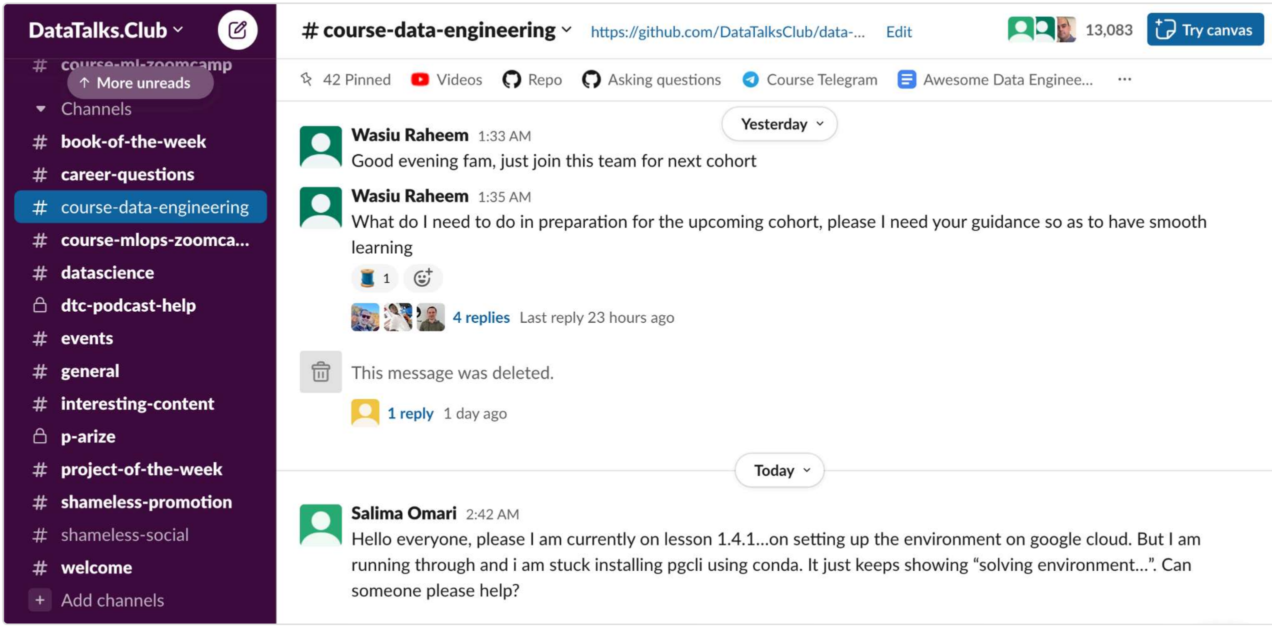


Data Engineering Zoomcamp certificate showing the certificate requirements

To earn your certificate, you must complete the course with a live cohort and fulfill three key requirements:

1. **Build a capstone project**: Create an end-to-end data pipeline that demonstrates your mastery of the course concepts
2. **Submit on time**: Meet the project submission deadline to qualify for certification
3. **Peer review**: Evaluate and provide feedback on 3 fellow students' projects during the peer review process

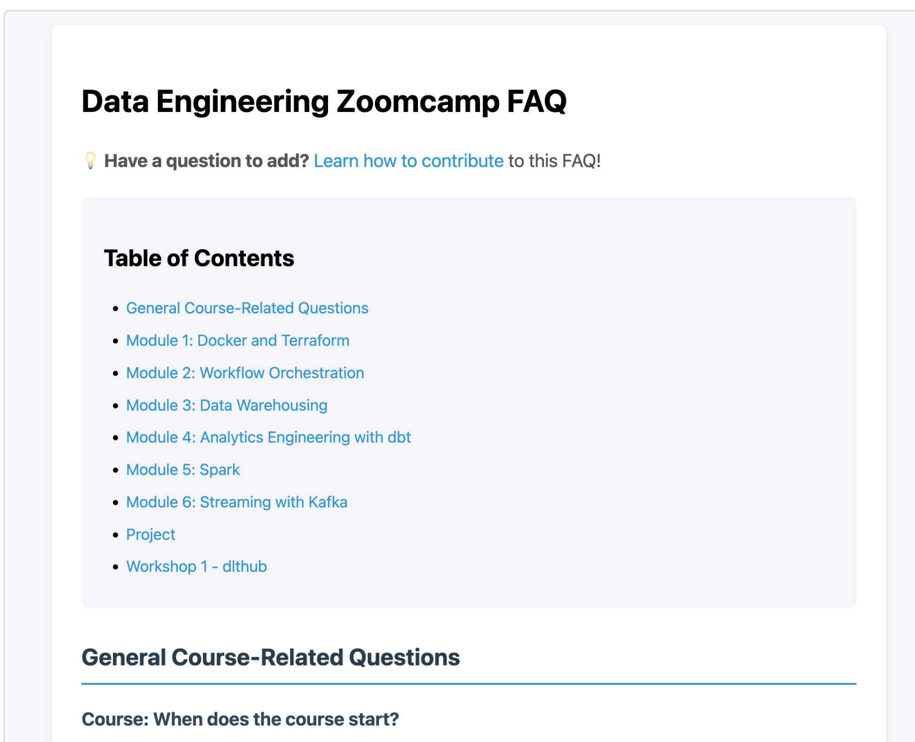
Data Professionals



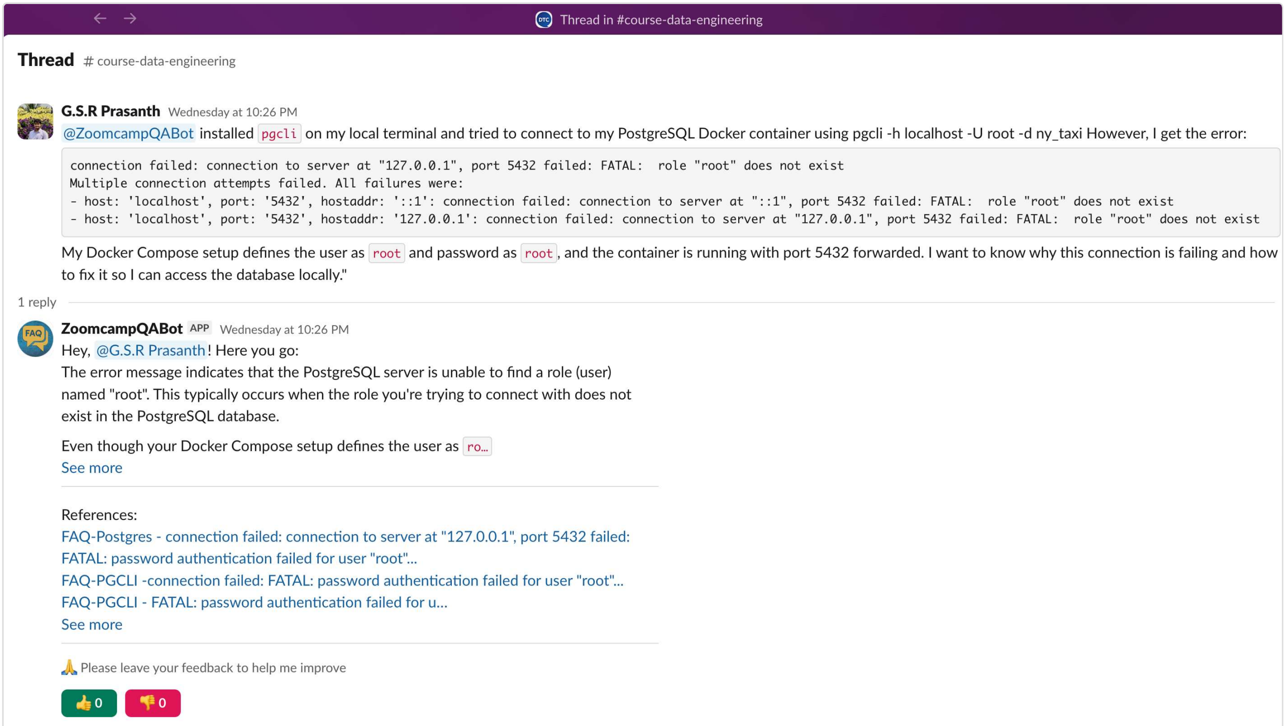
Active discussions and peer support in our dedicated Slack community channel

DataTalks.Club is a global community of 80,000+ data professionals who connect on Slack to share knowledge, ask career questions, and discuss everything from analytics and visualization to machine learning and data engineering. As one of the largest digital groups dedicated to data, it's where you'll find data scientists, ML engineers, data analysts, and enthusiasts at all career stages.

When you join a cohort, the dedicated course channel becomes your home base. Here, you'll troubleshoot problems with peers working through the same challenges and share your progress and insights.



DataTalks.Club FAQ repository showing common questions and technical issues



DataTalks.Club Zoomcamp QABot in Slack providing quick answers when you need them

Zoomcamp vs. Bootcamp: What’s the Difference?

We often get asked what the difference is between the Data Engineering Zoomcamp and paid bootcamps.

Below, we list the key features of the Data Engineering Zoomcamp and how they compare to paid bootcamps.

Feature	DE Zoomcamp (Cohort)	DE Zoomcamp (Self-paced)	Paid Bootcamps
Cost	Free	Free	$2,000–$10,000+
Format	9-week cohort with fixed schedule	Learn anytime at your own pace	Fixed schedule, instructor-led
Homework	Weekly scored assignments	Available but no scoring	Weekly with instructor feedback
Projects	Capstone project with peer review and scoring	Build on your own, no evaluation and scoring	Instructor-reviewed projects
Certificate	Yes, after completing project + peer reviews	No certificate	Certificate of completion
Community Support	Active Slack + optional live Q&A sessions	Slack community only	Instructor-led, 1:1 or group mentorship
Learning in Public	Encouraged with bonus points	Optional	Rarely emphasized
Timeline	9 weeks (Jan–Mar 2026)	Flexible, self-paced	Typically 12–24 weeks
Best For	Career switchers or experienced data engineers wanting community and accountability	Self-motivated learners exploring data engineering	Those needing intensive structured mentorship and guidance

To get started with the Zoomcamp, you can either join a live cohort or learn at your own pace.

All materials are freely available in the Data Engineering Zoomcamp GitHub repository. Each module has its own folder (e.g., `01-docker`, `03-data-warehouse`), and cohort-specific homework and deadlines are in the `cohorts` directory. Lectures are pre-recorded and available in our official Data Engineering Zoomcamp YouTube Playlist, so you can watch at your own pace.

Learn at Your Own Pace

The self-paced mode lets you start immediately and progress on your own schedule. You'll get access to the course materials and the community and can complete the course at a pace that works for you.

All you need is to go to the Data Engineering Zoomcamp GitHub repository and start learning. It serves as a central hub for the course for easier navigation through the course materials. All the lectures are pre-recorded and available on YouTube in our official Data Engineering Zoomcamp YouTube Playlist, so you can watch whenever it suits you.

You can also join the DataTalks.Club Slack community to get help and support from the community in the `#course-data-engineering` channel. Homework and solutions are available on the course platform, and you can build a project for your portfolio.

> Remember, self-paced learning does not include homework submissions, project evaluations, or the ability to earn a certificate. To receive certification, you need to join an active cohort.

Join a Live Cohort

> **2026 Cohort:** Starts January 2026. Register here: Fill in this form

When you join a live cohort, you'll work through the same materials as self-paced learners, but with the added structure of a published schedule and the energy of hundreds of peers progressing alongside you.

The course runs once per year (starts around January). Each module typically spans one week: you watch the lectures, complete hands-on exercises, and submit a homework assignment. Your submissions get scored and appear on an anonymous leaderboard.

After completing all six modules, the capstone project phase begins. You'll build your own end-to-end pipeline, submit it through our form, and peer-review at least three other students' projects while yours gets reviewed by your peers. This reciprocal process gives you valuable feedback on your work and exposes you to different approaches and solutions you might not have considered.

Even if you join after the official start date, you can still follow along — but note that some homework forms may already be closed. All active deadlines are listed on the course platform.

> To earn a certificate, you'll need enough time to complete one final project and the required peer reviews. Details are in the Projects and Certificate sections.

Ready to join DE Zoomcamp? Here's how it works:

1. Register for the course, you'll be automatically accepted into the next cohort
2. Join the DataTalks.Club Slack community and the `#course-data-engineering` channel for updates, questions, and peer support
3. (Optional) Get a head start by exploring the GitHub repository and watching lectures before the cohort officially begins
4. When the cohort starts, you'll receive an email with the full schedule and submission deadlines
5. Follow the weekly rhythm: watch lectures, complete exercises, submit homework
6. During the final three weeks, build and submit your capstone project, then peer-review three other projects
7. Receive your certificate once your project passes peer review

Comparison

We summarized the key differences between the two joining options in this table:

Feature	Self-Paced	Live Cohort
Timing	Learn at your own pace, start anytime	Fixed 9-week schedule (January–March each year)
Course Materials	Full access to GitHub repository and YouTube lectures	Full access to GitHub repository and YouTube lectures
Community	Access to Slack community (`#course-data-engineering`)	Access to Slack community (`#course-data-engineering`)
Homework	Available but not scored	Scored automatically, appears on leaderboard
Projects	Build on your own, no evaluation	Submit 1 final project (capstone) with peer review
Certificate	Not available	Available after completing project and peer reviews
Structure	Flexible, no deadlines	Weekly rhythm with deadlines and peer accountability

Join the next cohort →

Testimonials

We've collected some testimonials from our students who have completed the Data Engineering Zoomcamp.

> Thank you for what you do! The Data Engineering Zoomcamp gave me skills that helped me land my first tech job.
>
> — Tim Claytor (Source)

> Three months might seem like a long time, but the growth and learning during this period are truly remarkable. It was a great experience with a lot of learning, connecting with like-minded people from all around the world, and having fun. I must admit, this was really hard. But the feeling of accomplishment and learning made it all worthwhile. And I would do it again!
>
> — Nevenka Lukic (Source)

> One of the significant things I inferred from the Zoomcamp is to prioritize fundamentals and principles over ever-evolving tools and tech stacks. Hugely grateful to Alexey Grigorev for putting together this incredible course and offering it for free.
>
> — Siddhartha Gogoi (Source)

> Such a fun deep dive into data engineering, cloud automation, and orchestration. I learned so much along the way. Big shoutout to Alexey Grigorev and the DataTalksClub team for the opportunity and guidance throughout the 3 months of the free course.
>
> — Assitan NIARE (Source)

> If you're serious about breaking into data engineering, start here. The repo's structure, community, and hands-on focus make it unparalleled.
>
> — Wady Osama (Source)

For more details about how our courses work, check the Zoomcamp logistics guide. You can also find more information about this course in the Data Engineering Zoomcamp documentation.

Is the Data Engineering Zoomcamp a free data engineering course?

Whats the difference between a DataTalks.Club Zoomcamp and a typical data engineering bootcamp?

Does the Data Engineering Zoomcamp offer a certificate?

How does the Data Engineering Zoomcamp certificate work?

When does the next cohort of the DE Zoomcamp start?

Can I join the Data Engineering Zoomcamp 2025 cohort?

Who runs the Data Engineering Zoomcamp?

What are the prerequisites for the Data Engineering Zoomcamp?

How much time does it take?

Can I learn at my own pace?

What if I get stuck?

Where is the GitHub repository for the Data Engineering Zoomcamp?

Where can I find the Data Engineering Zoomcamp syllabus?

Is it the same as DE Zoomcamp, Data Engineer Zoomcamp, or Zoomcamp Data Engineering?

Is the Data Engineering Zoomcamp good preparation for data engineering jobs?

How do I join the DataTalks.Club Slack community?

Is there a Discord or Telegram channel for DataTalks.Club?

Join the next cohort →

Related Posts

Free DataTalks.Club Courses: ML, Data Engineering, MLOps, LLM & AI Dev Tools Zoomcamps

Earn certificates and gain practical experience in ML, data engineering, MLOps, LLMs, AI development tools, and stock market analytics

Read more →

ML Zoomcamp: Free Machine Learning Engineering Course and Certification

Master Machine Learning Engineering with Python in 4 Months

Read more →

MLOps Zoomcamp: Free MLOps Course and Certification

Learn to deploy, monitor, and maintain ML models in production with MLflow, Docker, AWS, and monitoring tools

Read more →

LLM Zoomcamp: Free LLM Engineering Course and Certification

Master LLM Engineering: Build Production-Ready AI Applications from Scratch

Read more →

AI Dev Tools Zoomcamp: Free Course to Master AI Tools for Developers

Learn how to integrate AI into real developer workflows, from AI coding assistants to agents, CI/CD, DevOps, and no-code automation.

Read more →

. . .